

Variable Definitions and Transformations

1. Study Variables

Dependent Variable

Reading Literacy Score

- Source: OECD PISA Database
- Description: Mean reading literacy performance score reported for each participating country and PISA cycle.

Independent Variables

Highest Parental Occupational Status (HISEI)

- Source: OECD PISA Database
- Description: Highest occupational status of parents.

Highest Parental Education in Years (PARED)

- Source: OECD PISA Database
- Description: Years of parental education.

Cultural Possessions (CULTPOS)

- Source: OECD PISA Database
- Description: Index reflecting cultural resources available at home.

Sense of Belonging (BELONG)

- Source: OECD PISA Database
- Description: Students' perceived sense of belonging at school.

2. Missing Data Treatment

Reading Literacy Scores

Missing values were identified for:

- Malaysia (2006)
- Peru (2006)

These values were estimated using linear interpolation based on adjacent PISA cycles.

Independent Variables

CULTPOS

The missing value for 2022 was estimated using linear extrapolation based on the trend observed between 2015 and 2018.

BELONG

Missing values for 2003, 2006, and 2009 were estimated using backward extrapolation based on the trend observed in subsequent cycles.

3. Data Standardization

All variables were standardized using the Z-score transformation:

$$Z = (X - \mu) / \sigma$$

where:

- X = observed value
- μ = variable mean
- σ = standard deviation

The standardized dataset is provided in:

06_standardized_model_dataset.xlsx

4. Sliding Window Transformation

A sliding window approach was applied to transform the time-series data into supervised learning sequences.

Window size = 3

Example:

Input:

$[t-2, t-1, t]$

Output:

$[t+1]$

This transformation was used for LSTM and GRU models.

5. Recursive Multi-Step Forecasting

Future values were generated recursively:

- 2025 forecast generated using observations up to 2022
- 2028 forecast generated using observations up to 2025
- 2031 forecast generated using observations up to 2028

6. Forecasting Models

The following forecasting models were implemented:

- ARIMA
- ARIMAX
- Prophet
- LSTM
- GRU

7. Evaluation Metrics

Model performance was evaluated using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Dynamic Time Warping (DTW)